$$P(y) = \int P(y|x)\, P(x)\, \mathrm{d}x$$

$$x_{n+1} = \lambda x_n (1 - x_n)$$

$$\lim_{x \to \infty} \frac{1}{1 + x^{-1}}$$

$$\sum_{n=1}^{N} n(n-1)$$

$$\int_b^a f(x)\, \mathrm{d}x$$

Introduction to LaTeX

Session 2: Document Structure, Formatting and Figures

1. What is LaTeX and why use it?
2. How does it work? – `.tex (+.bib)` → Compiler → `.pdf`
3. Document structure – `\documentclass{}` → preamble → `\begin{document}`
4. Basic document formatting – `\maketitle`,`\toc`,`\section{}`,...
5. Text formatting – `\bf`,`\it`,`\rm`,`\sc`,`\tt`,...
6. Commenting – % This is a comment
7. Diagnosing compilation errors

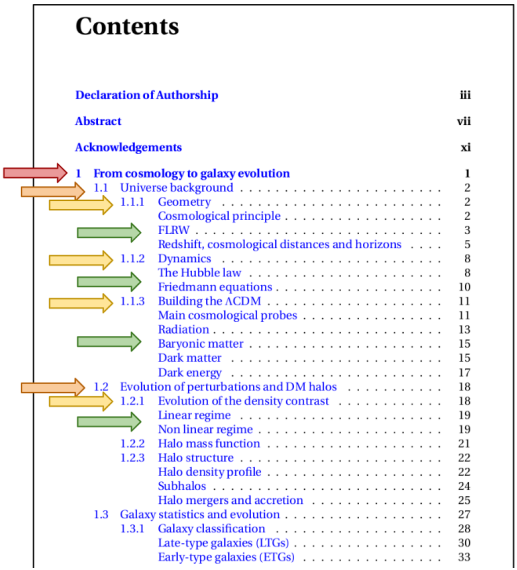
Outline

1. How to organise large documents
2. Using packages
3. Fiures
4. Referenceing sections, figures, equations, etc.

1 Organising large documents

Your thesis will be composed of:

- Chapters
- Sections
- Subsections
- Subsubsections



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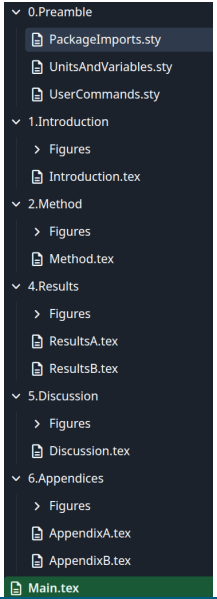
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Organising large documents can be done easily in LaTeX

The document can be broken down into sections and subsections. Each of these can have its own directory containing a `.tex` and figures.

Long preambles (user defined commands and/or imports) can be put in a `.sty` file that acts as a custom package.

These can then be added to the main document with `\include{}`, `\input{}`, or `\import{}{}`



There are three ways to add text from other files to the main document:

- `\input{filename}`
- `\include{filename}`
- `\import{path}{filename}`

Only the `Main.tex` should contain a preamble. The inserted `.tex` files will inherit this.

`\input{filename}`

- Inserts the contents of `filename.tex`
- LaTeX will not start a new page
- You can nest `\input` commands in files

`\include{filename}`

- Inserts the contents of `filename.tex`
- LaTeX will start a new page before processing `filename.tex`

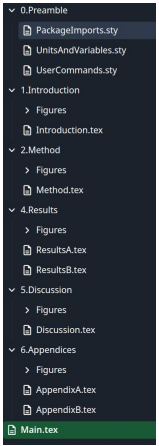
`\import{path}{filename}`

- `\input` and `\include` can be prone to errors when nested
- So, you could consider using `\import` from the `import` package

If your preamble is very long (e.g. a long list of user macros), these definitions / imports can be placed in a `.sty` file

- By starting the `.sty` with `\ProvidesPackage{MyMacros}`, you are telling LaTeX this is a package called MyMacros
- Then simply include this package in the preamble of `Main.tex` as `\usepackage{MyMacros}`

By following these steps you can easily compartmentalise large documents that are hundreds of pages.





```
% Document class
\documentclass{article}
% Preamble
\usepackage{graphicx}
\usepackage{lipsum}
\usepackage[margin=1in]{geometry}
\usepackage{amssymb}
% Custom packages
\usepackage{PackageImports}
\usepackage{UnitsAndVariables}
\usepackage{UserCommands}
% Document Information
\title{My Latex Document}
\author{Your Name}
\date{\today}

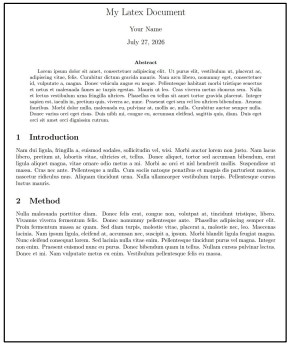

\begin{document}
\maketitle

    \begin{abstract}
        \lipsum[1]
    \end{abstract}

    \section{Introduction}\label{sec:Introduction}
        \import{./Introduction/}{Introduction.tex}

    \section{Method}\label{sec:Method}
        \import{./Method/}{Method.tex}

    \section{Results}\label{sec:Results}
        \import{./Results/}{Results.tex}
```



My Latex Document

Your Name

July 27, 2026

Abstract

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

1 Introduction

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

2 Method

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

2 Using Packages

Packages

Much of LaTeX's functionality is available in third-party packages.
Packages are included in the preamble with `\usepackage{name}`.
Some common packages are:

- `amsmath` – Useful maths package from the American Mathematical Society
- `graphicx` – Handles the inclusion of figures
- `hyperref` – Handles the inclusion and formatting of links
- `natbib` – bibliography manager
- `geometry` – Handles page formatting

A full list of packages is available on CTAN – `https://ctan.org/pkg/:c`

Custom packages

You can also create your own custom packages:

- Create a `.sty` file with the first line `\PackageProvides{PackageName}`. This tells LaTeX that it is a package.
- You can then load this package via `\usepackagePackageName`
- This can be useful if you have a lot of user defined commands (e.g. a library of units and variables) that you want in most you files

3

Including Figures

Including images

To add figures you need the package `graphicx`, included via

```
\usepackage{graphicx}
```

You can define the path to the figures directory in the preamble via

```
\graphicspath{{./path/to/figures/}}
```

The simplest inclusion of an image is done via

```
\includegraphics{figure.png}
```

A figure is typically inserted with the code block

```
\begin{figure}
  \centering
  \includegraphics[width=5cm]{fig.png}
  \caption{This is a caption...}
  \label{fig:KeyName}
\end{figure}
```

`\begin{figure*}` will span the page in a multi-column document.

A figure is typically inserted with the code block

```
\begin{figure} ⇐ Start the figure environment
  \centering
  \includegraphics[width=5cm]{fig.png}
  \caption{This is a caption...}
  \label{fig:KeyName}
\end{figure}
```

`\begin{figure*}` will span the page in a multi-column document.

A figure is typically inserted with the code block

```
\begin{figure}
  \centering ⇐ Centre the image
  \includegraphics[width=5cm]{fig.png}
  \caption{This is a caption...}
  \label{fig:KeyName}
\end{figure}
```

`\begin{figure*}` will span the page in a multi-column document.

A figure is typically inserted with the code block

```
\begin{figure}
  \centering
  \includegraphics[width=5cm]{fig.png} ⇐ Insert Fig.
  \caption{This is a caption...}
  \label{fig:KeyName}
\end{figure}
```

`\begin{figure*}` will span the page in a multi-column document.

A figure is typically inserted with the code block

```
\begin{figure}
  \centering
  \includegraphics[width=5cm]{fig.png}
  \caption{This is a caption...} ⇐ Add a caption
  \label{fig:KeyName}
\end{figure}
```

`\begin{figure*}` will span the page in a multi-column document.

A figure is typically inserted with the code block

```
\begin{figure}
  \centering
  \includegraphics[width=5cm]{fig.png}
  \caption{This is a caption...}
  \label{fig:KeyName} ⇐ Label the figure
\end{figure}
```

`\begin{figure*}` will span the page in a multi-column document.

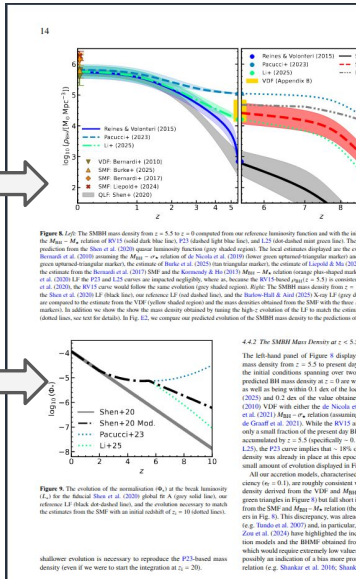
A figure is typically inserted with the code block

```
\begin{figure}
  \centering
  \includegraphics[width=5cm]{fig.png}
  \caption{This is a caption...}
  \label{fig:KeyName}
\end{figure}
```
⇐ **Close the figure environment**

`\begin{figure*}` will span the page in a multi-column document.

```
% figure* example
\begin{figure}
    \centering
    \includegraphics[width=0.9\textwidth]{Soltan.png}
    \caption{{\emph Left:} The SMBH mass density…}
    \label{fig:SoltanArg}
\end{figure}
```

```
% figure example
\begin{figure}
    \centering
    \includegraphics[width=0.45\textwidth]{LFNorm.png}
    \caption{The evolution of the normalisation…}
    \label{fig:LFNorm}
\end{figure}
```



```
\begin{figure}
  \centering
  \begin{subfigure}{0.45\textwidth}
    \centering
    \includegraphics[width=\textwidth]{figA.png}
    \caption{This is a caption...}
    \label{fig:KeyName}
  \end{subfigure}
  \hfill
  \begin{subfigure}{0.45\textwidth}
    \centering
    \includegraphics[width=\textwidth]{figA.png}
    \caption{This is a caption...}
    \label{fig:KeyName}
  \end{subfigure}
\end{figure}
```

Subfigure 1
(Same format as a normal figure)

\hfill
add some space between the panels

Subfigure 2
(Same format as a normal figure)

You can set the size of an image in the `\includegraphics` options

```
\includegraphics[width=5cm,height=4cm]{figure.png}
```

You can set either:

- absolute sizes (e.g. `5cm`)
- relative sizes (e.g. `0.5\textwidth` or `0.5\linewidth`)

You can set the alignment of an image in the `\includegraphics` options. E.g.

```
\includegraphics[width=5cm,height=4cm,left]{figure.png}
```

will align the image to the left. Other options are `right`, `inner`, `outer`, etc.

- LaTeX willautomatically position figures at the top, bottom, or in a dedicated page
- LaTeX aims to minimise white space, prevent bad page breaks, and ensure it is near (ish) to the relevant text
- his is generally okay, but may not be ideal if there's many figures, or large figures

- **h** – Place here, i.e. approximately the same place as the source text
- **t** – Place at the top of the page
- **b** – Place at the bottom of the page
- **p** – Place on a page for floats only
- **!** – Override LaTeX's definition of "good" float placement
- **H** – Place exactly HERE. Approximately equivalent to **h!**

Note that journals will often disable the float options in their templates.

4

Referencing Figures

(as well as sections, equations, ...)

All Figures (as well as Sections, Tables, equations, etc.) should be given a label when they are defined

```
\label{fig:name}
```

Within the environment. It is common for the label to be formatted as `fig:[name]` to make for easy searching.

You can then simply reference the Figure in text as

... as seen in Figure `\ref{fig:name}`, the ...

(Similarly, section are labelled as `\label{sec:name}`, subsections as `\label{ssec:name}`, tables as `\label{tab:name}`, appendices as `\label{app:name}`, equations as `\label{eq:name}`.)

All Figures (as well as Sections, Tables, equations, etc.) should be given a label when they are defined

```
\label{fig:name}
```

Within the environment. It is common for the label to be formatted as `fig:[name]` to make for easy searching.

You can then simply reference the Figure in text as

... as seen in Figure `\ref{fig:name}`, the ...

(Similarly, section are labelled as `\label{sec:name}`, subsections as `\label{ssec:name}`, tables as `\label{tab:name}`, appendices as `\label{app:name}`, equations as `\label{eq:name}`.)

5

Hands-on Exercise

To your document from last session:

- Upload and insert figures
- Resize figures
- Create captions
- Reference figures in the text
- Split the document into multiple files

6

Homework

Produce a two-page report with two figures and cross-references.



Extra Slides